U-I-490B
June 29, 1977
SUPERSEDING
Int. Fed. Spec.
U-I-00490a(DSA-DM)
February 24,1969 and
Fed. Spec. U-I-490
January 3, 1957

FEDERAL SPECIFICATION

IMPRESSION MATERIAL (MODELING COMPOUND), DENTAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental impression material (modeling-compound) for use in taking impressions and for forming trays to be used in the oral cavity.

1.2 Classification. Impression material shall be of the following colors and forms as specified (see 3.3.2 and 6.1):

Color A	-	Green
Form 1	-	Sticks
Color B	-	Grey
Form 1	-	Sticks
Color C	-	Red
Form 1	-	Sticks
Form 2	-	Cakes

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

FSC 6520

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN DENTAL ASSOCIATION:

ADA Specification No. 3 for Dental Impression Compound.

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, IL 60611.)

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3. REQUIREMENTS

3.1 Material. The impression material (modeling compound) essentially shall be a composition of resins and fillers with necessary additives such as plasticizers and coloring agents.

3.2 Description. Impression material shall be thermoplastic impression compound used to take preliminary impressions of oral structures and to form a tray to be used with other types of impression materials. Impression material shall have firm smooth margins after trimming with a knife at room temperature (23 ± 2 degrees Centigrade (°C)).

3.2.1 Colors and forms. Color A (green) and color B (grey) impression material shall be furnished in sticks (form 1) each measuring 5/16 ± 1/16 inch in diameter and 4-1/2 ± 1/4 inches in length. Color C, red (brick red) impression material shall be furnished in either cakes (form 2) conforming to figure 4 for shape and dimensions, or sticks of the dimensions specified above. Color A, green impression compound shall be uniform and suitable for use in corrected impression techniques. Colors B and C impression compound shall be uniform and suitable for taking impressions in the oral cavity.

3.2.2 Unit of issue. Unless otherwise specified (see 6.1) impression material shall be furnished in the applicable unit of issue as listed in table I:

TABLE I. Unit of issue

Color and form	Unit of issue
Color A, Form 1 (Green, sticks)	Box containing a minimum of 1/4 pound of impression material in stick form.
Color B, Form 1 (Grey, sticks)	Box containing a minimum of 1/4 pound of impression material in stick form.
Color C, Form 1 (Red, sticks)	Box containing a minimum of 1/4 pound of impression material in stick form.
Color C, Form 2 (Red, cakes)	Box containing a minimum of 1/2 pound of impression compound in cake form.

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3.3 Physical properties.

3.3.1 Odor and flavor. The impression compound shall have no unpleasant odor or flavor when tested as specified in 4.3.1.

3.3.2 Working range. The impression compound shall soften to a workable consistency within the temperature range specified in table II for the applicable color, when tested as specified in 4.3.2:

TABLE II. Working range

Color	Working range
A, Green	122° to 124°F (50° to 51°C)
B, Grey	128° to 130°F (53° to 54°C)
C, Red	132° to 133°F (55.5° to 56.0°C)

The working time of the impression compound shall be from 60 to 80 seconds when tested as specified in 4.3.2.

3.3.3 Flow. The percent of flow shall be as specified in table I when tested as specified in 4.3.3.

TABLE III. Flow test

Temperature	Flow (percent)	
Centigrade	Colors B and C	Color A
37.0° 45.0°	6.0 (MAX) 88.0 (Min)	6.0 (MAX) 88.0 (Min)

3.3.4 Surface detail (impression test). Colors A, B, and C impression compound shall accurately record the complete surface detail of the test block (see figure 1) at 45°C when tested as specified in 4.3.4.

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3.4 Toxicity. Prior to delivery, the manufacturer shall furnish a statement that the impression compound, when used according to the directions supplied, has no toxic effect on a normal healthy person. This statement shall be sent directly to the Technical Services Branch (Code ATTS-2), Directorate of Medical Materiel, Defense Personnel Support Center, 2800 S. 20th Street, Philadelphia, PA 19101.

3.5 Certification. Suppliers shall furnish at time of award a dated certificate and data showing that the impression compound they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 3 for Dental Impression Compound, type I. Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

3.6 Instructions. Instructions, including method of softening, working temperature, and curve or data showing the shrinkage of the compound from 40°C to 20°C shall be supplied with each package. In addition, the contractor shall mail two copies of the instructions to the contracting officer for distribution.

3.7 Shelf life. The shelf life of the modeling compound shall be a minimum of 24 months from date of manufacture, and no less than 20 months shall remain from date of delivery to the Government, when stored at 59° to 86°F (15° to 30°C). The inspection test date shall be 24 months from date of manufacture.

3.8 Lot or batch number. Each unit container shall be marked with a lot or batch number and date of manufacture.

3.9 Workmanship. Workmanship shall be first class throughout. Impression compound shall be free from defects which may detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Certificates of compliance. Certificates of compliance, supplied by the manufacturer of the component or raw material, may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.1.3 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Impression material shall be examined for defects including, but not limited to, those listed in table IV.

TABLE IV. Classification of defects

Categories	Defects
Major:	
101	Type not as specified.
102	Form not as specified.
103	Color not as specified.
104	Presence of foreign materials such as dirt or lumpy material.

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TABLE IV. Classification of defects (cont'd)

Categories	Defects
Minor A:	
201	More than one cake or more than three sticks broken.
202	Instructions not complete or not applicable to material furnished.
203	Packaging, packing, or marking not as specified.
204	Shape of cakes not as specified.
Minor B:	
251	Weight not as specified for applicable type, form, and color.
252	Dimensions of sticks not as specified.

The inspection levels and acceptable quality levels shall be as listed in table V. The unit of product for examination shall be one box (unit of issue) of impression compound.

TABLE V. Examination criteria

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Testing. Impression compound shall be tested as specified in table VI and 4.3. Sufficient number of cakes or sticks shall be drawn at random from each of two boxes per lot or control number. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE VI. Test criteria

Characteristic	Requirement	Test procedure	Number of determinations per lot or batch number
Odor and flavor	3.3.1	4.3.1	1 1/
Working range	3.3.2	4.3.2	3 1/
Flow	3.3.3	4.3.3	2 2/
Surface detail (impression test)	3.3.4	4.3.4	3 1/
Toxicity	3.4	4.3.5	1 1/

1/ A lot shall be rejected if any determination fails to comply with the test requirement.

2/ Result shall be the average of 2 specimens.

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4.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing amd marking requirements of section 5. Defects shall be scored as specified in table VII. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE VII. Examination of preparation for delivery

Examination	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.2.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636 or PPP-B-676, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the appropriate box specification.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Odor and flavor. The pleasantness or unpleasantness of the odor and flavor shall be determined by smelling and tasting the material.

4.3.2 Working range test. The test shall be conducted in a 23°C ± 2°C room. A water bath shall be adjusted to and maintained at the working range temperature of the impression compound (see 3.3.2). A 30 gm ± 2 gm sample of the impression compound shall be used. Place sample in the water bath and warm for a period of 15 minutes. Remove warmed sample from water bath and work (knead) in wet hand for 15 seconds. Return sample to water bath for an additional 5 minutes conditioning. Remove sample from water bath again and work

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4.3.2 Working range test. (cont'd)

(knead) in wet hand until the sample loses plasticity, or proper working consistency, and can no longer be molded by finger pressure. The total elapsed time from final removal from the water bath to the loss of plasticity shall be termed working time. Three tests shall be performed on each lot (control) number of compound. All three determinations shall pass the test for acceptability of the lot (control) number of the compound involved.

4.3.3 Flow test.

4.3.3.1 Preparation of specimens. A sufficient number of cakes or sticks of compound shall be broken into pieces and placed in an iron crucible of 50 milliliter (ml) capacity. The crucible shall be placed 130 millimeters (mm) below a 250-watt infrared lamp bulb. The stirred compound shall be heated to a temperature of 75° ± 5°C (checked with a thermometer) until melted throughout and then shall be poured into a mold (figure 2). The mold shall be a flat, stainless steel metal plate 6.0 mm thick, containing four holes 10.0 mm in diameter. The mold shall be preheated to a temperature of 55° ± 5°C and placed on a smooth glass slab (152 mm long, 76 mm wide, 19 mm thick) which shall have been preheated to the same temperature. After the mold has been overfilled with material, a smooth, flat, tinfoil-covered glass plate, preheated to 55° ± 5°C, shall be placed on top of the mold. A load of 9,000 grams (gm) shall be applied to the top of the tin-foil glass plate for 10 minutes. Then the mold shall be placed in water at a temperature of 10°C for 10 minutes. The upper tin-foiled plate shall then be removed and the excess material shall be trimmed away. This can be accomplished by using a metal scraper with the top surface of the mold as a guide. The mold may then be removed from the glass slab by gently tapping the side of the mold. The specimens shall be removed from the mold by chilling in water for 10 minutes at 10°C, and shall be stored at room temperature, 23° ± 2°C, for 24 hours before testing.

4.3.3.2 Method of test. The original length of the specimen shall be determined using a metric micrometer caliper. Four measurements shall be made around the circumference and one measurement shall be made on the center of specimen. The measurements shall be recorded to the nearest 0.005 mm. The specimen and flow testing instrument (figure 3) shall be placed in a water bath and held at the testing temperature for 20 minutes prior to testing. The flow testing instrument as illustrated consists of the following: A, a weight, B, a nonconducting shaft; and C, a brass platen. The total weight in air of these three components is 2,000 gm. The weight A shall be separated a minimum of 76 mm from the brass

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platen on the shaft. This shaft shall be of hard rubber or of a similar poor thermal conductor to avoid loss of heat due to conduction. The diameter of the brass platen which touches the specimen shall be no less than 51 mm. The thickness of the brass platen shall not exceed 6.5 mm. The temperature in the bath shall be controlled to within ± 0.1°C of the required temperature. A thin sheet of waterproof cellophane shall be placed between the instrument and each end of the specimen. The bottom of the specimen shall be 51 mm below the surface of the water in the bath. A constant axial load of 2,000 gm shall then be applied to the specimen for 10 minutes, after which the specimen shall be removed and cooled in air to room temperature. The cellophane shall be stripped off, and the final length shall be determined in the same way as the original length. The flow, as evidenced by the change in length, shall be reported as percentage of the original length. The value for flow at any temperature shall be the average value for two specimens from each lot (control) number and shall be reported to the nearest 0.1 percent.

4.3.4 Surface detail (impression test).

4.3.4.1 Preparation of impression specimen. The impression specimen shall be a disk 40 mm in diameter and between 4 mm and 7 mm in thickness. The cake of compound shall be warmed in water and the disk cut. If the cake is thinner than 4 mm, two cakes shall be flamed and pressed together before cutting the disk. Three specimens shall be tested from each lot (control) number. All three specimens shall pass the test.

4.3.4.2 Method of test. A water bath shall be adjusted to 45.0° ± 0.1°C. A flat plate (to support the specimen while it is coming to the required temperature); a test block (figure 1); and a brass weight, having a base 50 mm in diameter and weighing 1,000 ± 5 gm in air, shall be placed in the bath and allowed to come to the temperature of the bath. (The water level in the bath shall be 30 ± 5 mm above the top surface of the test block). The impression specimen shall then be placed in the water bath on the flat plate. Twenty minutes later, the specimen shall be centered on the test block, a sheet of waterproof cellophane shall be placed over the specimen, and the weight shall be placed on the specimen. Ten minutes later, the weight shall be removed, the test block and specimen shall be taken from the bath and chilled at a temperature of 10°C, and the specimen shall be removed from the test block. The specimen shall be examined to determine if the ridges corresponding to the large cross grooves (B, figure 1) of the test block are sharp, and the ridges corresponding to the fine grooves (C, figure 1) are complete and visible to the unaided eye for at least 30 mm of their length.

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4.3.5 Toxicity. The material, prepared according to the manufacturer's instructions and used as an impression material, shall not cause visible evidence of irritation of the oral mucosa.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. One quarter pound of color A, B or C, form 1 or one-half pound of color A, form 2 shall be packaged in a box conforming to PPP-B-676. Closure of the box shall be as specified in the appendix of the box specification.

5.1.2 Level C. Impression material shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. One hundred unit packages containing form 1 or 2 of one color only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, banding and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. One hundred unit packages containing form 1 or 2 of one color only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged impression material shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

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5.3 Marking.

5.3.1 Civil agencies.

5.3.1.1 Interior packages and exterior containers. In addition to any marking specified in the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military.

5.3.2.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown. Type II shelf life marking as specified in MIL-STD-129 shall be shown.

5.3.2.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129. Type II shelf life markings as specified in MIL-STD-129 shall also be shown.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color and form required (see 1.2).
(c) If unit of issue other than that specified (see 3.2.2) is desired.
(d) If toxicity statement is to be furnished to another Government activity (see 3.4).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation
6520-00-536-2710	IMPRESSION MATERIAL, DENTAL, Cakes, Red, 1/2 lb.	Color C, Form 2
6520-00-765-0251	IMPRESSION MATERIAL, DENTAL, Sticks, Gray, 1/4 lb.	Color B, Form 1

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NSN	Item Identification	Specification Designation
6520-01-020-1427	IMPRESSION MATERIAL, DENTAL, Sticks, Green, 1/4 lb.	Color A, Form 1
6520-01-020-1428	IMPRESSION MATERIAL, DENTAL, Sticks, Red, 1/4 lb.	Color C, Form 1

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6520-1432

Civil Agency Coordinating Activities:
COM-NBS
GSA-FSS
HEW-HSM
HEW-FDA

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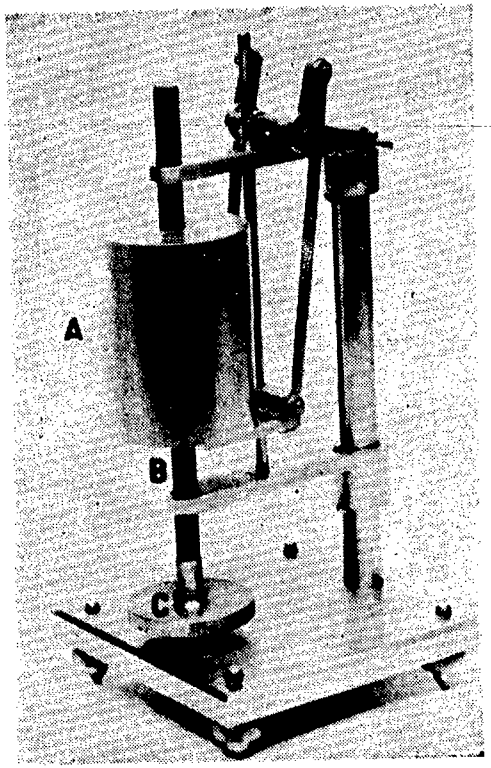


FIGURE 3.—Flow testing instrument. A; weight B; nonconducting shaft. C; platen.

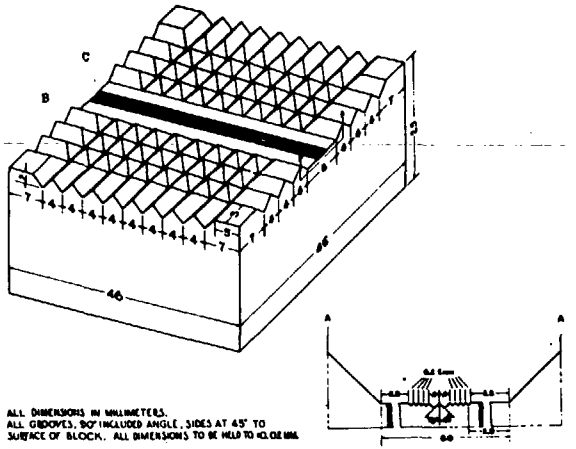


FIGURE 1.—Test block for dental impression compounds.

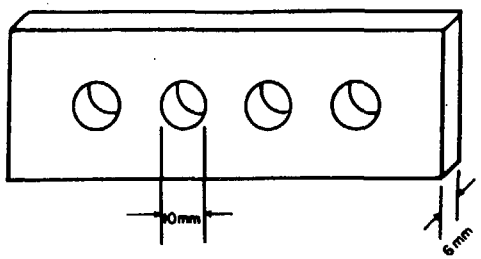
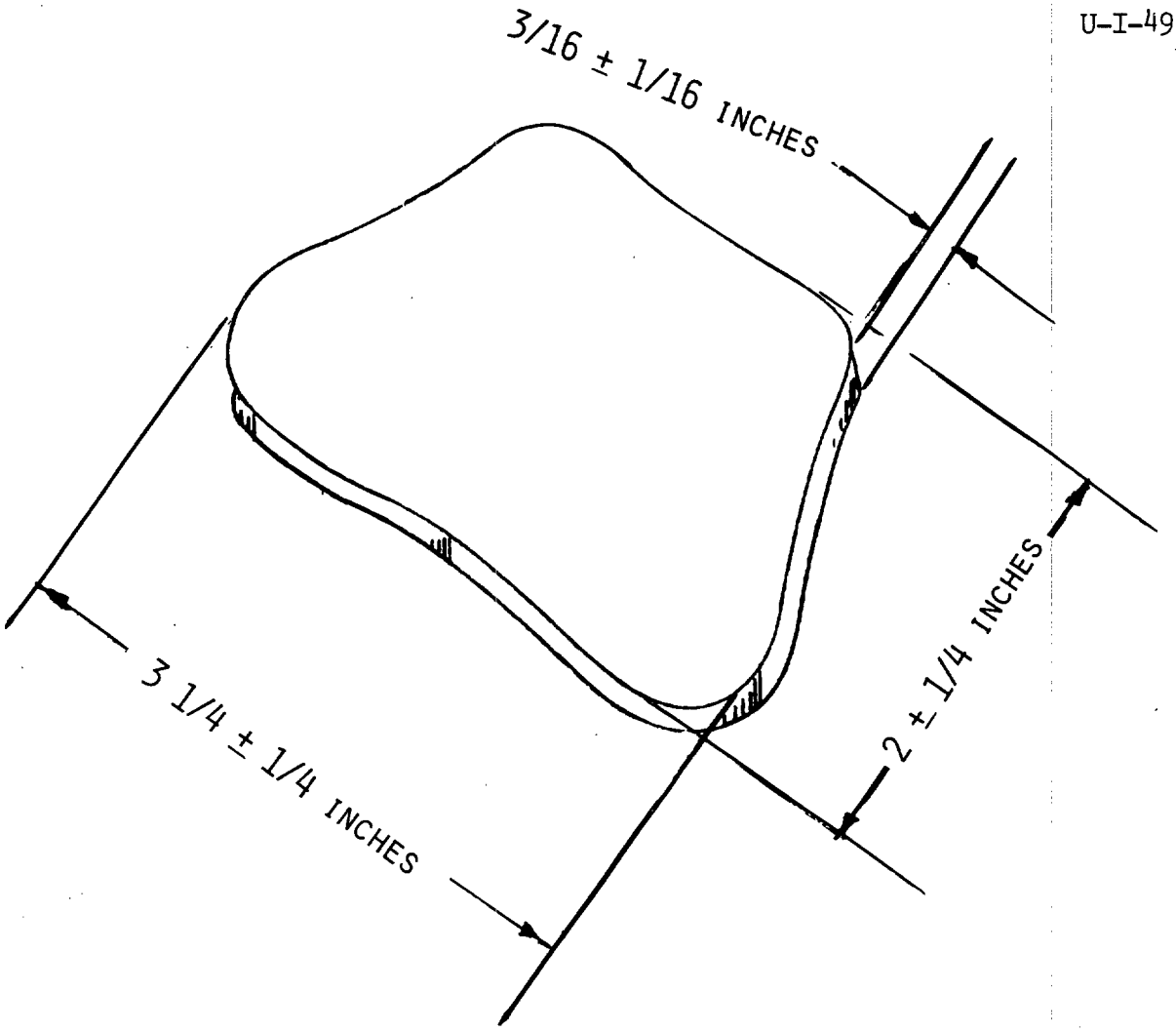


FIGURE 2.—Mold for forming flow specimens.

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CAKE FORM

FIGURE 4

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 45 cents each.

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DOCUMENT IDENTIFIER AND TITLE
U-I-490B IMPRESSION MATERIAL (MODELING COMPOUND), DENTAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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